BIPRO ROY

AI-Driven Learning Analyst | Machine Learning, NLP & Learning Analytics
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Professional Summary

Final-semester Computer Science and Engineering student at AIUB who builds AI-powered learning tools and the analytics behind them. Developed FluentUp, an English-learning app that places learners with an assessment, adapts their path, schedules review with spaced repetition and tracks their progress, with a Gemini-based AI tutor. Experienced in fine-tuning Transformer models for Bangla NLP and in comparing machine learning and statistical forecasting models, with papers at ICIS 2025 and IEEE TENSYMP 2026. Looking to use machine learning, NLP and data analysis to measure and improve how people learn.

Technical Skills

- **Programming:** Python, TypeScript, JavaScript, SQL, C++
- **Machine Learning & NLP:** Scikit-learn, Hugging Face Transformers (DistilBERT fine-tuning), PyTorch, LSTM, SVR, ARIMA, Prophet, text preprocessing, regex-based information extraction, stemming, inverted indexing
- **LLM Integration:** Gemini API, prompting for structured JSON output, response validation with Pydantic, rule-based fallbacks
- **Data Analysis & Visualization:** Pandas, NumPy, Recharts dashboards, time-series model evaluation (RMSE, MAE, MAPE), classification metrics
- **Learning Analytics:** placement assessment, spaced repetition (SM-2), learner progress tracking (XP, streaks, lesson completion), mistake tracking
- **Development & Tools:** FastAPI, Streamlit, Next.js, React Native (Expo), PostgreSQL, SQLAlchemy, Docker, Git, GitHub Actions, pytest, Playwright

Education

B.Sc. in Computer Science and Engineering | American International University-Bangladesh (AIUB), Dhaka | 2023 – Expected April 2027

- CGPA: 3.40 / 4.00 (final semester)
- Relevant coursework: Machine Learning, Natural Language Processing, Data Structures and Algorithms, Database Management Systems, Web Technologies

Higher Secondary Certificate (Science) | Gaokhali School and College | 2022 | GPA 5.00 / 5.00

Relevant Projects

FluentUp – AI-Powered English Learning App | React Native, TypeScript, FastAPI, PostgreSQL, Gemini API | github.com/roybipro/fluentup

- Built an English-learning app for Bengali-speaking learners in which a weighted 5-question placement assessment recommends a starting CEFR or IELTS learning path.
- Designed a learning engine with 6 levelled paths, lesson prerequisites, 6 quiz types and a mistake bank that stores each learner's errors for targeted review.
- Implemented the SM-2 spaced-repetition algorithm to schedule vocabulary review from each learner's recall performance.
- Built a progress dashboard that tracks XP, current and longest streaks, completed lessons and achievements.
- Integrated the Gemini API on the server for an AI conversation tutor, grammar feedback and IELTS answer evaluation, validating every response with Pydantic and falling back to rule-based checks when the API is unavailable.
- Wrote 60 backend (pytest) and 18 mobile (Jest) tests; secured accounts with Supabase Auth and PostgreSQL row-level security.

Bangla Fake News Detector | Python, Hugging Face Transformers, PyTorch, Streamlit | github.com/roybipro/fake-news-project

- Fine-tuned multilingual DistilBERT to classify Bangla news articles as authentic or fake from combined headline and body text.
- Handled class imbalance (about 7,000 authentic vs. 1,000 fake articles) by undersampling to a balanced training set.
- Evaluated on a held-out test set with accuracy, precision, recall and F1.
- Deployed an interactive Streamlit app for single-article predictions.

CareerLab – Career and Research Tracker | FastAPI, PostgreSQL, Next.js, TypeScript | Live: careerlab-opal.vercel.app

- Built a full-stack workspace for tracking job applications, research opportunities and research projects, with Kanban boards, deadlines and reminders.
- Created an analytics page with status charts and monthly application trends computed from the database.
- Covered the system with 239 backend tests against a real PostgreSQL database and Playwright end-to-end tests.

AIUB Faculty Data Collector (NLP coursework) | Python, regular expressions | github.com/roybipro/NLP-Faculty-Data-Scraper-AIUB

- Extracted names, e-mails and room numbers for 197 faculty members using regular expressions only, after locating the JSON feed behind a JavaScript-rendered page.
- Built stemming, an inverted index of 641 stems and a regex query tool over faculty research-interest text.

Research Publications

- **Roy, B.** "AI for Stock Market Forecasting in Emerging Economies." 7th International Conference on Integrated Sciences (ICIS 2025), Dhaka, Bangladesh, 25 October 2025. Presented. Supervisor: Md. Mortuza Ahmmed. aiub.edu/Files/student-research/AI_for_Stock_Market_Forecasting_in_Emerging_Economies.html
- **Roy, B.** (co-author). "Development of a Cloud-Linked Low-Cost Machine Learning-Enabled Weather Monitoring System." 2026 IEEE Region 10 Symposium (TENSYMP 2026). Published in IEEE Xplore. DOI: 10.1109/TENSYMP68909.2026.11704037

Research Experience

Undergraduate Researcher | American International University-Bangladesh | 2025

- Compared LSTM, Support Vector Regression, Prophet and ARIMA for forecasting Bangladeshi bank share prices using Dhaka Stock Exchange, Yahoo Finance and Kaggle data.
- Evaluated models with RMSE, MAE and MAPE; ARIMA achieved the lowest test error (RMSE 0.91, MAE 0.64, MAPE 1.17%).
- Presented the findings at ICIS 2025 under the supervision of Md. Mortuza Ahmmed.

Research interests: Natural Language Processing, Bangla NLP, Large Language Models, LLM reliability and hallucination detection, AI in education

Languages

Bengali (native), English (professional working proficiency)